Pritom Bhowmik

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Research Interests

Trustworthy AI agents and adversarial machine learning: memory and retrieval poisoning in LLM agents, the security–utility cost of defenses, trust between agents in multi-agent systems, and human–AI interaction.

Education

M.S., Data Science — May 2025
Montclair State University — *Montclair, NJ*
GPA: 3.97 / 4.00 · Alpha Epsilon Lambda Honor Society
Focus: multi-agent retrieval architectures and tool calling

B.Tech., Computer Science — Jul 2020
Institute of Engineering and Management — *Kolkata, India*
Coursework: algorithms, data structures, software engineering, databases, computer networks

Publications

Preprints and Manuscripts

[1] **Pritom Bhowmik**. *The Price of Safety: Benign-Case Utility and Token Overhead of Memory-Poisoning Defenses in LLM Agents*. arXiv:2609.22818 [cs.CR], 2026. Submitted, Workshop on Agentic Engineering (AGENT) at ICSE 2027.

[2] **Pritom Bhowmik** and Hao Liu. *Outcome-Bound Trust: Falsifiable Claims as a Security Primitive for Memory in Agent-to-Agent AI Systems*. To be submitted, Workshop on Trustworthy Evolution of AI Agents at AAAI 2027. Code.

Earlier Work

[3] **Pritom Bhowmik**. *Machine Learning in Production: From Experimented ML Model to System*. ScienceOpen Preprints, 2022. doi:10.14293/S2199-1006.1.SOR-.PPKHXND.v1

[4] **Pritom Bhowmik**. *Vulnerability of Neural Networks to Adversarial Attack and Defenses in Computer Vision*. IEEE Computer Society, Bangladesh Chapter, 2021. Abstract.

Research Experience

Independent Researcher — Sep 2025 – Present
Memory Security in LLM Agent Systems
Study what defending agent memory costs and how agents can trust memory they share [1, 2].

- Showed on the LoCoMo benchmark that write-time memory defenses carry no detectable utility cost, while a read-time reranker cuts accuracy by 4.4 points and, on attack-free traffic, wrongly flags 9.9% of the memories it retrieves, removing memory from 45% of answers [1].
- Designed Outcome-Bound Trust, which turns agent messages into checkable claims and caps what an agent commits to each counterparty by the claims it has already honored; in a supply-chain simulation it cut losses from lying agents by 98.1% at a 5.43% utility cost, with its safety invariants model-checked in TLA+ [2].

Graduate Researcher, Data Science Lab — Jan 2024 – May 2025
Montclair State University — *Montclair, NJ*
Advisor: Dr. Hao Liu
Studied how a multi-agent RAG system should route and combine retrieval across heterogeneous sources for pharmaceutical supply-chain decisions (master's project).

- Designed a routing protocol that classifies each query as factual, analytical, or hybrid and dispatches it to retrieval, analytics, or web-search agents over inventory, transport, and regulatory data.
- Built a six-agent RAG system with hybrid BM25 and dense retrieval over 15+ databases, raising retrieval accuracy from 72% to 97% and cutting query latency from 3.2 minutes to 12 seconds.

Teaching Experience

Adjunct Instructor, Computer Science Sep 2025 – Jul 2026
Monroe University *New Rochelle, NY*

- Taught undergraduate and graduate courses: Programming Logic, Python, Database Systems, and Data Science.
- Designed Python, C++, and SQL lab curricula for 100+ students per semester; mentored undergraduate and graduate students.

Industry Experience

Product Management Intern, Data & AI Jun 2025 – Dec 2025
Sentari AI *New York, NY*

- Analyzed 200+ user sessions and designed experiments to raise engagement with LLM-generated prompts; turned product requirements into technical specifications for the AI engineering team.

Data Analyst (Freelance) Jun 2021 – Jul 2023
Upwork *Remote*

- Designed A/B tests and SQL analyses of user behavior for client products, raising conversion by 31.4%.

Technology Project Management Intern Jan 2021 – Apr 2021
Upkey · IE Business School capstone *Remote*

- Built an ML model that segments cancer patients by support needs to route them to relevant services.

Technical Skills

Research Methods	Controlled ablations, bootstrap confidence intervals, McNemar's test, inter-rater agreement, LLM-as-judge evaluation, formal verification (TLA+ model checking), property-based testing
Programming	Python, C++, SQL
LLMs & Agents	LangChain, LangGraph, FAISS, hybrid retrieval, RAG, A2A protocol, Ollama (local models); OpenAI, Anthropic, Gemini, DeepSeek, and xAI APIs
Machine Learning	TensorFlow, scikit-learn, XGBoost, pandas, NumPy
Tools	LaTeX, Git, Docker, AWS, Flask